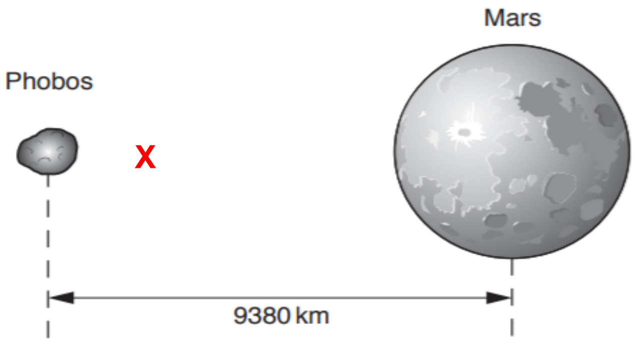
1 **(a)** **(i)** Nearer to Phobos [1]



(ii) Gravitational force provides centripetal force [1]

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$ [1]

$$M = \frac{v^2 r}{G} = \frac{(2.14 \times 10^3)^2 \times (9380 \times 10^3)}{6.67 \times 10^{-11}}$$ [1]

$= 6.44 \times 10^{23}$ kg [1]

(b) **(i)** The g-field of the Earth and Mars are pointing in <u>opposite directions</u> [1]

the magnitude of the Earth's field is greater than that of Mars. The sum of the 2 fields gives a non-symmetrical shape. [1]

(ii) At the point where the net force / field is zero, r = 4.4×10^{10} from the Earth.

Distance from Mars = $5.8 \times 10^{10} - 4.4 \times 10^{10} = 1.4 \times 10^{10}$m [1]

$$\frac{GM_E}{r_e^2} = \frac{GM_M}{r_m^2}$$

$$\frac{M_E}{M_m} = \left(\frac{4.4 \times 10^{10}}{1.4 \times 10^{10}}\right)^2 = 9.88$$ [1]

(iii) Min energy = mass × min change in potential [1]

Min change in potential = area from Mars to zero point of g. [1]

Min energy = 20 square × $(0.1\times10^{10})(0.2\times10^{-6})$ × 200 kg
= 200 × 53000
= 8.0×10^5 J [1]

2 **(a)** (at very low pressure) gas occupies very large volume / particles are further apart